# Topic 5 — Soils

**Covers syllabus (click to expand)**

**Formation & profile:** Soil Formation | Soil Profile | Soil Classification **Types:** Alluvial (जलोढ़) | Black (Regur (रेगुर)) | Red | Laterite (लेटराइट) | Desert | Mountain | Saline (लवणीय) | Alkaline | Problem Soils (Acidic, Saline, Alkaline) | Peaty (पीट)/Marshy (ICAR set) | Karewa (करेवा) **Management:** Soil Texture | Soil Fertility | Soil Erosion | Soil Conservation | Soil Conservation Methods | Jhum / badlands **UP Focus:** Bhabar (भाबर)–Tarai–Alluvial–Bundelkhand–Vindhyan belts. Local names include Mar, Kabar, Parua and Rakar. Learn Usar separately.

**Sources baked in:** NCERT Geography Class 11 (Soils), Class 12 (Resources), soil types + Indian formation factors + erosion (Kullar), **Ghatnachakra** World Geography **The Soil** (CB–197+), UPPCS Prelims PYQs 2018–2025 **Weight:** ★★★ — type–region–crop, laterite Fe/Al, black ≠ Himalaya, khadar/bangar, usar remedies, conservation methods **Last verified:** August 2026 **Current Affairs:** Soil Health Card → RKVY **Soil Health and Fertility**; **>25 crore** cards (Jul 2025); slogan *Swasth Dharaa, Khet Haraa*

## Current Affairs (this topic)

| Year | Fact | Why asked | Source |
|---|---|---|---|
| **19 Feb 2015** | Soil Health Card launched at **Suratgarh, Rajasthan**; slogan **Swasth Dharaa, Khet Haraa**; 2015 = International Year of Soils | Scheme–place–slogan | PIB |
| **2022–23 onward** | SHC merged into RKVY as **Soil Health and Fertility** | New name vs old scheme | PIB / DA&FW |
| **Jul 2025** | **>25 crore** Soil Health Cards distributed; card reports **12 parameters** (N,P,K,S, Zn,Fe,Cu,Mn,B, pH, EC, OC) | Numbers / parameters | PIB Aug 2025 |
| Ongoing | **WDC–PMKSY 2.0** = watershed soil–water conservation (not a soil-type topic) | Scheme vs conservation method | Jal Shakti (शक्ति) / DA&FW |
| Static | World Soil Day (विश्व मृदा दिवस) **5 December** (UN;) | Date trap vs 5 June | UN / PYQ |

## N.1 Formation, Profile, Classification

**Pedogenesis** is soil formation — rock weathers, organics mix in, and horizons appear over time.

- Ancient Indian names called fertile soil **Urvara** and sterile soil **Usara**.

- First scientific soil classification is linked to **V.V. Dokuchaev**; the five-factor model is **Jenny's CLORPT**.
- Soil is loosely mineral particles + humus + water + air + organisms (regolith's plant-supporting top).

### How rock becomes soil (weathering)

Think of weathering as three jobs that turn parent rock into loose material for soil:

| Type | Simple meaning | What changes? | India note |
|---|---|---|---|
| **Physical** | Breaks rock into **smaller pieces** | Size only — chemistry stays the same | Heat cracking, frost, wind abrasion on exposed rock |
| **Chemical** | **Changes minerals** inside the rock | New compounds form (hydration, oxidation, carbonation, solution) | Strong under **hot wet monsoon** toward laterite |
| **Biological** | Life **mixes and dissolves** | Roots split rock; organic acids + burrowers loosen and enrich | Forest floors, earthworms |

Physical weathering **crushes** rock. Chemical weathering **transforms** minerals. Biological weathering **mixes** and dissolves. All three usually work together.

### Residual vs transported

- **Residual** (mostly peninsular) soils form **in place** on the rock beneath them. They are also called **sedentary** soils.
- Regur on **Deccan Trap basalt** and red soil (लाल मिट्टी) on **granite–gneiss** are residual.
- **Transported** (extra-peninsular plains) soils are carried in by rivers, wind, or ice — **alluvium** is the classic. These are often tagged **azonal** (weak horizons, little relation to the rock underfoot).
- Same climate on different rock can still give different soil. **Basalt** weathers to **black** soil. **Granite–gneiss** weathers to **red** soil.
- Farming, irrigation, and deforestation can strip a profile **faster** than nature rebuilds it.

### Parent rock in Indian conditions

Indian parent material is commonly grouped into five families. Climate can still override the rock.

- **Ancient crystalline and metamorphic** rocks (granite, gneiss, schist) are the peninsular basement. They weather to **red** soils because of iron oxide.
- **Cuddapah and Vindhyan** sedimentary rocks weather to **calcareous and clayey** soils. They are mostly **poor in metalliferous** minerals.
- **Gondwana** sedimentary rocks give comparatively **immature, uniform, low-fertility** soils.
- **Deccan basalts** (about ten lakh km² of trap) are rich in titanium, magnetite, aluminium and magnesium. They weather to dark, moisture-holding **regur** (रेगुर).
- **Tertiary and Mesozoic sediments** of the plains and Himalaya give **porous, immature alluvium**. The soil has little memory of a single parent rock.
- Peninsular soils are generally **coarser** and closer to the rock. Plain soils are generally **finer** and more fertile.

### CLORPT

| Factor | Role | India example |
|---|---|---|
| **C**limate | Weathering + leaching intensity | Laterite on hot-wet Ghats |
| **L**iving organisms | Humus, mixing, nutrient cycling | Forest cover (वन आवरण) in the Himalaya |
| **R**elief | Depth, drainage, erosion | Thin soils on steep Himalayan slopes |
| **P**arent material | Mineral base of the soil | Deccan Trap weathers to regur |
| **T**ime | How mature the horizons are | Mature black vs young khadar |

- Steep slopes give **thin** soils; valleys give **deeper** soils. Chambal ravines (चंबल) and bare Himalayan south faces show how erosion **stops** a profile from forming.
- Colour alone does **not** prove fertility.
- **Edaphic** means the **soil** factor for crops/vegetation — not climatic, not biotic, not topography as the term.

#### Climate can override parent rock

- The **same granite** can give **laterite** in a wet monsoon belt and a **non-laterite** soil in a drier belt.
- **Hot summer and low rainfall** can make **black** soil even in parts of **Tamil Nadu** that sit on gneiss, not trap.
- In **Rajasthan** (थार), both granite and sandstone weather to **sandy** soil under arid climate.
- Arid evaporation exceeds rainfall. Vegetation is thin, humus is low, and colour is light. Lime accumulates, so the soil is **pedocal**.
- Cold Himalayan climate slows decay, so forest soils there are often **acidic**.

#### Why laterite looks red and bakes like brick

- Heavy rain **leaches** bases and silica downward.
- In the dry season, evaporation exceeds rain. **Capillary rise** brings iron and aluminium oxides to the surface, so the soil looks red.
- Alternate wet and dry plus a blazing sun **bakes** the top so hard it resembles brick. That is the Latin *later* fact.

#### Vegetation

- Leaf fall adds humus. Densely forested tracts often hold some of India's **best** hill soils.
- Vegetation type and soil type travel together in India.

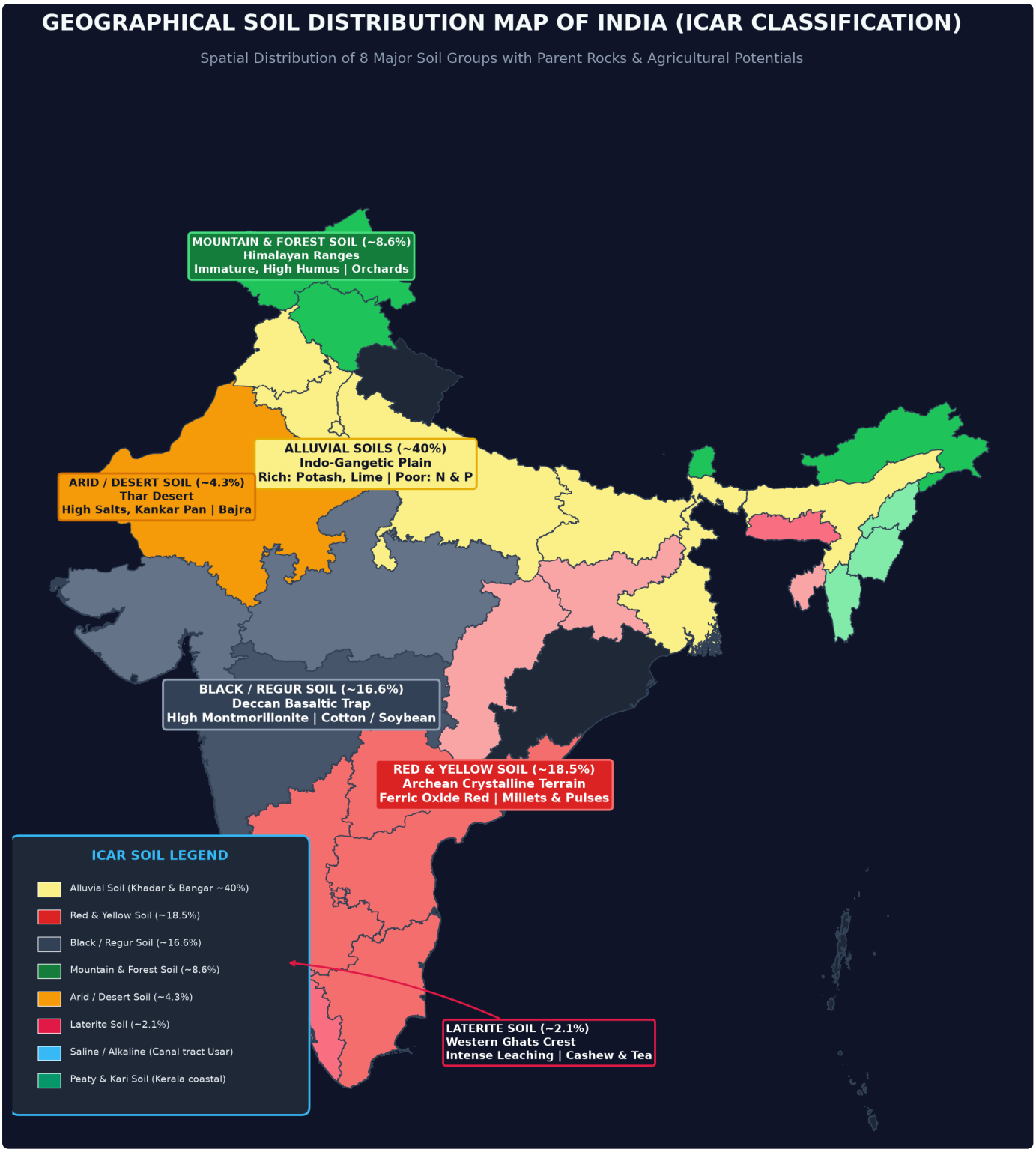
#### Profile (top to bottom)

| Horizon | Content | tag |
|---|---|---|
| **O** | Organic litter / humus | Forest floors |
| **A** | Topsoil: mineral + humus | Crop root zone |

| Horizon | Content | tag |
|---|---|---|
| **E** | Leached zone (if present) | **Eluviation** (wash **out**) |
| **B** | Clay / Fe / Al / lime accumulation | **Illuviation** (wash **in**) / subsoil |
| **C** | Weathered parent, not fully soil | Transition |
| **R** | Unweathered bedrock | Base |

- Young alluvium (**khadar**) often has **weak** horizons.
- Mature residual soils show a **clear** O–A–B–C–R sequence.
- Ploughing mixes O and A.

## ICAR / school classification




Notes use this set, not full USDA taxonomy names. Read each row for how it forms, where it lies, the star crop, and the classic trap.

| Soil | How it forms | Core belt | Star crop / use | Classic trap |
|---|---|---|---|---|
| **Alluvial** | River-borne silt (**transported**) | Indo-Gangetic, Brahmaputra (ब्रह्मपुत्र), E-coast deltas (NCERT ~**40%**; other notes ~**46%**) | Wheat, rice, sugarcane, jute | UP is **alluvial**, not a black-soil state |

| Soil | How it forms | Core belt | Star crop / use | Classic trap |
|---|---|---|---|---|
| **Black (Regur)** | On **Deccan Trap basalt** (residual) | MH–MP–GJ–N KA–TG | **Cotton** | **Not** Himalayan |
| **Red & Yellow** | On **granite–gneiss** (crystalline) | TN, KA, S MH, Odisha, Chota Nagpur | Millets, pulses | Red ≠ laterite; not always barren |
| **Laterite** | **Hot-wet** intense leaching | Ghats, NE hills, plateau edges (**Malabar**/Kerala is the classic home belt) | Cashew, **tapioca**, tea, coffee, cinchona | **Rich** in Fe/Al — not poor; not RJ/UP plains |
| **Desert / Arid** | Arid climate + sand | W Rajasthan (Thar) | Millets if irrigated | Not forever barren |
| **Mountain / Forest** | Altitude + slope | Himalaya, NE hills | Orchards, terraces | Thin on slopes; ≠ regur |
| **Saline / Usara** | Salts + poor drainage | Canal plains, arid belts, coastal deltas, **Rann of Kutch** | Needs reclamation | More water **without** drainage worsens it |
| **Peaty / Marshy** | Waterlogged organic pile-up | Kerala (**Kuttanad**), Sundarbans, N Bihar, S Uttarakhand (उत्तराखंड) | Drainage first | Black colour ≠ cotton regur |

**World overlay:** Brazil's **Selva** is equatorial rainforest. **Terra Roxa** is fertile dark basaltic coffee soil. It parallels fertile volcanic soils and is **not** Indian regur.

### Soil ↔ state match bank (classic Prelims)

> **Prelims Anchor Rule:** Commission match questions ("Match List-I with List-II") test the **core geographic anchor** (the epicenter state where the soil type is most prominent or defining), not an exclusive claim that the state possesses only that single soil type.
>
> **10-Second Elimination Trick:** Always lock the two most rigid pairs first: 1. **Desert / Arid** ↔ **Rajasthan** (or **Black / Regur** ↔ **Maharashtra**). 2. Then check the 4 code options — in >85% of UP-PCS / UPSC questions, locking these two eliminates 3 wrong choices immediately without needing to debate borderline red vs alluvial borders!

### 1. Pan-India ICAR Soil Groups ↔ State Match Bank

| Soil Group (ICAR) | NCERT Area Share | Primary Match State (Commission Anchor) | Alternate / Secondary States (If primary absent) | Why this association? (Geological / Climatic basis) | Classic Prelims Trap / Misconception |
|---|---|---|---|---|---|
| **Alluvial Soil** *(जलोढ़ मृदा)* | **~40%** (Class 11) ~46% (other) | **Uttar Pradesh** (उत्तर प्रदेश) | Bihar, West Bengal, Punjab, Haryana, Coastal deltas (AP/Odisha/TN) | Extensive Indo-Gangetic river floodplains; transported silt (*khadar* & *bangar*) | **Trap:** Bundelkhand (southern UP) is **not** alluvial (it has black *Mar/Kabar* & red *Rakar/Parua*); but UP as a whole holds the single largest alluvial tract (~90% of state area). |
| **Black Soil / Regur** *(काली / रेगुर मृदा)* | **~15–16%** | **Maharashtra** (महाराष्ट्र) | Madhya Pradesh (Malwa), Gujarat (Kathiawar), North Karnataka, Telangana | Formed *in situ* by weathering of **Deccan Trap basaltic lava**; rich in titaniferous magnetite; high clay & moisture retention | **Trap:** Regur is strictly peninsular volcanic; **never** match with Himalaya or Gangetic plains. Do not mistake local black patches in eastern India (Rajmahal) as main Deccan regur. |
| **Red & Yellow Soil** *(लाल एवं पीली मृदा)* | **~18.5%** *(2nd largest)* | **Andhra Pradesh** (आंध्र प्रदेश) / **Tamil Nadu** | Odisha, Chhattisgarh, Chotanagpur (Jharkhand), Southern Karnataka | Formed on **ancient crystalline & metamorphic granite–gneiss** rocks under low-to-medium rainfall; red from ferric oxide ($Fe_2O_3$) diffusion | **Trap:** Red soil is **not** laterite! Red soil has good tilth and responds well to fertilizers (pulses, millets, oilseeds); laterite is intensely leached, acidic, and bakes rock-hard. |

| Soil Group (ICAR) | NCERT Area Share | Primary Match State (Commission Anchor) | Alternate / Secondary States (If primary absent) | Why this association? (Geological / Climatic basis) | Classic Prelims Trap / Misconception |
|---|---|---|---|---|---|
| **Laterite Soil** *(लेटराइट मृदा)* | **~3.7–4.3%** | **Kerala** (केरल - Malabar Coast) | **Maharashtra** (Western Ghats crests/Konkan if Kerala absent), Karnataka, Meghalaya (Shillong plateau) | Intense **tropical leaching** under alternate wet & dry seasons; silica & lime washed down (*eluviation*), Fe & Al oxides left on top; bakes like brick (*later*) | **Trap:** First identified in **Angadipuram, Kerala (Malabar)** by Francis Buchanan (1807). If Kerala is not in the options, look for **Maharashtra / Karnataka** (Western Ghats). Never pick northern alluvial plains (UP/Punjab) or desert (Rajasthan). |
| **Desert / Arid Soil** *(मरुस्थलीय मृदा)* | **~4.3–4.4%** | **Rajasthan** (राजस्थान - Western/Thar) | Northern Gujarat (Kutch fringe), Southern Haryana & Punjab | Hyper-arid to semi-arid climate, mechanical weathering; sandy texture, alkaline pH, high soluble salts, thick subsoil *kankar* (calcareous) pan | **Trap:** Not permanently sterile — highly productive when irrigated (Indira Gandhi Canal proved this for wheat/mustard/cotton). Do not match with wet NE states like Meghalaya! |
| **Forest & Mountain Soil** *(पर्वतीय / वन मृदा)* | **~8.6%** | **Jammu & Kashmir / Ladakh / Himachal Pradesh / Uttarakhand** | Sikkim, Arunachal Pradesh, higher slopes of Western & Eastern Ghats | Formed under forest cover on steep mountain slopes; thin, skeletal, immature profile (*azonal*); acidic with slow humus decomposition in cold climates | **Trap:** Soil depth varies sharply by relief — thin on steep slopes, deep and fertile in valley bottoms (Kashmir, Doon, Kullu valleys). |

| Soil Group (ICAR) | NCERT Area Share | Primary Match State (Commission Anchor) | Alternate / Secondary States (If primary absent) | Why this association? (Geological / Climatic basis) | Classic Prelims Trap / Misconception |
|---|---|---|---|---|---|
| **Saline & Alkaline Soil** *(लवणीय एवं क्षारीय / ऊसर / रेह)* | **~2.1%** | **Uttar Pradesh** (Western & Central canal tracts) / **Gujarat** (Rann) | Punjab, Haryana, Coastal saline tracts (Sundarbans, Chilika) | High evaporation exceeding rainfall + canal over-irrigation without underground drainage → capillary rise of sodium, calcium & magnesium salts | **Trap: Gypsum ($CaSO_4 \cdot 2H_2O$)** is used for alkaline/sodic soils (replaces exchangeable Na with Ca); **Lime ($CaCO_3$)** is for acidic soils (low pH). Never use lime on alkaline/usar soils! |
| **Peaty & Marshy Soil** *(पीट एवं दलदली मृदा)* | Localized (~**1%**) | **Kerala** (Kuttanad & Alappuzha/Kottayam - local **Kari** soil) | Sundarbans (West Bengal), Coastal Odisha, Northern Bihar (Koshi floodland), Tarai (Uttarakhand) | Waterlogged conditions in humid tracts with poor drainage; slow anaerobic breakdown of dead vegetation → massive organic accumulation (humus **40–50%**) | **Trap:** Peaty soil is heavy and jet black, but it is **not** regur/black cotton soil! Regur is dry/semi-arid basaltic alkaline soil; Peaty is submerged, acidic, and organic-heavy. |

### 2. Special Micro-Regional Matches (High-Frequency Prelims Traps)

Examiners love inserting local, non-standard soil pockets into matching options to trip candidates relying solely on broad state names:

| Local / Regional Formation | Exact Match Location | Geological / Pedological Fact | Commission Trap to Avoid |
|---|---|---|---|
| **Karewa Soil** *(करेवा)* | **Jammu & Kashmir** (Kashmir Valley & Bhadarwah) | Thick lacustrine (lake-bed) silt & clay deposits from Pleistocene glaciation; famous for **Zafran (Saffron)**, almond, and walnut orchards. | Not river alluvium; not volcanic. It is specifically **lacustrine** (झील निक्षेप). |

| Local / Regional Formation | Exact Match Location | Geological / Pedological Fact | Commission Trap to Avoid |
|---|---|---|---|
| **Jharkhand Black Soil Pocket** | **Rajmahal Hills** (Santhal Pargana, Jharkhand) | Derived from localized weathering of **Rajmahal basaltic traps** (Jurassic volcanism), forming black clayey patches. | **Major Trap:** Do **not** pick the Damodar or Subarnarekha valley as black soil! Damodar valley is a Gondwana sedimentary rift basin (sandstone/shale/coal), not basalt. |
| **Matasi Soil** *(मटासी)* | **Chhattisgarh** (Mahanadi Basin plains) | Yellowish-red sandy-loam soil on slightly elevated drained uplands; rich in silica/iron; traditional staple soil for paddy. | Regional state-exam fact (CGPCS / UPPCS); part of the *Bhata → Matasi → Dorsa → Kanhar* toposequence, not a separate ICAR class. |
| **Bhabar vs Tarai Belt** | **Sub-Himalayan Foothills** (Piedmont strip from Indus to Teesta) | **Bhabar:** Porous pebble-gravel-boulder strip along Shiwalik foothills where mountain streams vanish underground (coarse, dry surface, unsuited for farming).<br>**Tarai:** Immediately south of Bhabar; damp, marshy, fine-silted waterlogged belt where lost streams re-emerge; rich in nitrogen & organic matter, excellent for sugarcane and paddy. | Do not invert their spatial order: **Shiwalik → Bhabar (north) → Tarai (south)**. Bhabar streams disappear; Tarai streams re-emerge. |

### 3. UP-Specific Soil Match Bank (UP-PCS & RO/ARO Special)

In UPPCS Prelims and RO/ARO, questions routinely test local nomenclature used in Bundelkhand and the Vindhyan plateau:

| Local Name | Region in UP | Core Soil Group | Distinguishing Characteristics & Crop Fit |
|---|---|---|---|
| **Mar (मार)** | Bundelkhand (Jalaun, Jhansi, Hamirpur) | **Black Soil** (Regur variant) | Highly clayey, deep black, extremely sticky when wet and cracks wide upon drying (*self-ploughing*); rich in iron & lime; prime for rabi gram, wheat, and oilseeds. |
| **Kabar (काबर)** | Bundelkhand (Banda, Hamirpur, Mahoba) | **Black Soil** (Coarse/compact variant) | Dark brown to blackish-grey; contains more sand than Mar; quickly hardens into unworkable stone-like clods if moisture is lost; highly sensitive to plowing window. |
| **Parua / Padwa (परुआ)** | Bundelkhand (Hamirpur, Jalaun, Yamuna ravines) | **Red Soil** (Sandy-loam variant) | Light reddish to yellowish-brown sandy loam; well-aerated with good drainage; low organic matter; highly productive when provided with light irrigation (mustard, wheat, jowar). |

| Local Name | Region in UP | Core Soil Group | Distinguishing Characteristics & Crop Fit |
|---|---|---|---|
| **Rakar (राकर)** | Bundelkhand (Sloping terrain, ravines, hill bases) | **Red Soil** (Gravelly/eroded variant) | Coarse, shallow, gravelly red soil formed on eroded granite knolls; low fertility, poor water retention; classified into *Moti Rakar* (gravelly) and *Patli Rakar* (fine); suited only for coarse millets (*bajra*, *sesamum*). |
| **Bhonta / Monta (भोंटा)** | Vindhyan tracts (Mirzapur, Sonbhadra, Prayagraj trans-Yamuna) | **Vindhyan Red Sandy Soil** | Coarse, reddish-brown stony soil formed from Vindhyan sandstone and shale weathering; low fertility, drought-prone; supports dry shrubs, pulses, and rainfed coarse grains. |
| **Matiyar / Dhub (मटियार)** | Ganga-Yamuna Doab & Eastern UP | **Alluvial Clay-Loam** | Fine clayey-loam alluvium with high moisture-retention capacity; prime paddy, wheat, and sugarcane belt. |

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### 4. Prelims Exam Drill: Classic Match Question Walkthrough

**PYQ Simulation — Match List-I (Soil Type) with List-II (State / Region):**

| List-I (Soil Type) | List-II (State / Region) |
|---|---|
| A. Alluvial soil | 1. Western Rajasthan |
| B. Black (Regur) soil | 2. Malabar coast (Kerala) |
| C. Laterite soil | 3. Uttar Pradesh |
| D. Desert soil | 4. Maharashtra |

**Codes:** - **(a)** A-3, B-4, C-2, D-1 - **(b)** A-4, B-3, C-1, D-2 - **(c)** A-3, B-1, C-4, D-2 - **(d)** A-1, B-4, C-2, D-3

**Click to view Answer & 15-Second Elimination Logic**

**Correct Answer:** **(a)** (A-3, B-4, C-2, D-1) **Step-by-step Elimination in the Exam Hall:** 1. **D (Desert soil)** is unequivocally **1 (Western Rajasthan)** → Look at the last number in the codes: only option **(a)** ends with **D-1**! Solved in 3 seconds. 2. **Double check:** **B (Black soil)** matches **4 (Maharashtra)**, **A (Alluvial)** matches **3 (Uttar Pradesh)**, and **C (Laterite)** matches **2 (Malabar coast, Kerala)**. 3. **Trap avoided:** If Maharashtra was paired with Laterite and Madhya Pradesh was in List-II for Black, you would assign MP to Black and Maharashtra to Laterite. But when Maharashtra and UP/Kerala/Rajasthan are present together, the standard ICAR archetype anchors are fixed as above.

### Stem to type (decision map)

> **How to Use the Decision Map:** In civil services prelims (UP-PCS, UPSC, BPSC, MPPSC), question stems frequently present distinctive pedological fingerprints or diagnostic phrases. When you

> identify a specific **trigger keyword** in the question stem, navigate directly to the verified soil type or problem condition.

### 1. Core Physical & Pedological Diagnostic Triggers

| Diagnostic Clue / Trigger Keyword in Stem | Target Soil / Formation | Key Scientific Rationale & Mechanism | Examiner's Trap / Distractor |
|---|---|---|---|
| **"Self-ploughing"** *(स्वतः जुताई)* / deep polygonal fissures when dry, swelling & unctuous/sticky when wet | **Black Soil / Regur** *(काली / रेगुर मृदा)* | High content of **montmorillonite** clay mineral; swells substantially on hydration, shrinks and cracks wide on drying, causing natural soil churning. | Not alluvial clay; not peaty soil. |
| **"High moisture retention"** / retains subsoil moisture for months; sustains rainfed rabi crops long after rains cease | **Black Soil / Regur** *(काली / रेगुर मृदा)* | Extremely fine, tight argillaceous texture and high capillary water-holding capacity. | Do not pick Alluvial or Red soil (both drain significantly faster). |
| **"Bakes like brick"** upon exposure to sun and drying / alternate wet-and-dry tropical leaching | **Laterite Soil** *(लेटराइट मृदा)* | Latin *later* = brick; intense tropical leaching washes away silica and bases (*desilication*), leaving an indurated residual crust of iron and aluminium oxides. | **Trap:** Red soil looks red from iron diffusion, but does **not** bake into rock-hard building bricks. |
| **"River-borne transported sediments"** / *khadar* (newer, light, annual flood silt) vs *bangar* (older, clayey, terrace alluvium) | **Alluvial Soil** *(जलोढ़ मृदा)* | Deposited by Himalayan and Peninsular rivers; azonal soil covering ~40% (NCERT) to ~46% of India's plain area. | UP and Bihar are overwhelmingly alluvial; do not assume all plain soils are homogeneous bangar. |
| **"Kankar nodules"** *(कंकड़ / Calcareous nodules)* in lower horizons / impure calcium carbonate concretions | **Old Alluvium (Bangar)** OR **Arid Soil** | Downward leaching and precipitation of calcium carbonate ($CaCO_3$) in subsoil under sub-humid to arid regimes. | *Khadar* (new alluvium) is completely devoid of kankar; kankar is characteristic of *Bangar*. |
| **"Thick sandy texture + high soluble salts + impermeable subsoil kankar layer"** | **Desert / Arid Soil** *(मरुस्थलीय मृदा)* | Mechanical disintegration dominates over chemical decay under arid conditions; high evaporation draws salts upward; kankar layer restricts root penetration. | Not permanently barren; highly productive when irrigated (e.g., Indira Gandhi Canal command area). |
| **"Thin, skeletal, immature profile on steep slopes"** / highly vulnerable to sheet erosion | **Mountain & Forest Soil** *(पर्वतीय मृदा)* | Relief and slope prevent horizon maturation; azonal/skeletal; acidic with raw un-decomposed humus in snow-bound belts; fertile in valley bottoms. | Not regur; not transported plains alluvium. |

| Diagnostic Clue / Trigger Keyword in Stem | Target Soil / Formation | Key Scientific Rationale & Mechanism | Examiner's Trap / Distractor |
|---|---|---|---|
| **"Waterlogged organic mass"** / high humus (40–50%) / dark, heavy, acidic / submerged lowlands | **Peaty & Marshy Soil** *(पीट एवं दलदली मृदा)* | Anaerobic conditions in humid wetlands prevent bacterial oxidation of dead plant matter (e.g., Kuttanad 'Kari' in Kerala). | **Trap:** Dark/black color is from decaying organic humus, **not** basaltic volcanic ash or regur titaniferous magnetite! |

### 2. Chemical Fingerprints & Remediation Triggers

| Diagnostic Clue / Chemical Trait in Stem | Soil Type / Problem Class | Required Soil Amendment / Remedy | Examiner's Trap |
|---|---|---|---|
| **White salt efflorescence / crust** on surface (*Reh, Kallar, Thur, Luni, Chopan*) | **Saline Soil** *(लवणीय मृदा / Solonchak)* | Leaching with fresh water + installing underground sub-surface tile drainage; salt-tolerant crops (barley, cotton). | Do not add lime or chemical precipitants; physical flushing and drainage are essential. |
| **High exchangeable sodium ($Na^+$) / pH > 8.5** / black alkali spots (*Usar*) | **Alkaline / Sodic Soil** *(क्षारीय मृदा / Solonetz)* | **Gypsum ($CaSO_4 \cdot 2H_2O$)** or Iron Pyrites ($FeS_2$) to replace exchangeable $Na^+$ with $Ca^{2+}$. | **Major Trap:** Adding Lime ($CaCO_3$) worsens alkaline soil! Lime is exclusively for acidic soils. |
| **Low soil pH (< 5.5) / high acidity** / strong leaching of bases | **Acidic Soil** *(अम्लीय मृदा - Laterite / Hill soils)* | **Agricultural Lime** (Quicklime CaO, Slaked lime $Ca(OH)_2$, or Limestone $CaCO_3$), Dolomite, or Wood Ash. | Never prescribe gypsum for acidic soils; gypsum does not raise soil pH significantly. |
| **Rich in Potash and Lime, but universally deficient in Nitrogen, Phosphorus & Humus** | **Alluvial & Black Soils of India** | Supplement with NPK fertilizers (ideal Indian ratio **4:2:1**); green manuring with *Dhaincha* or *Sunnhemp*. | Almost all Indian soils (except peaty/forest) are inherently deficient in **Nitrogen, Phosphorus, and Humus**. |
| **Rich in Iron & Aluminium, severely depleted in Silica, Lime, Magnesia, and Potash** | **Laterite Soil** *(लेटराइट मृदा)* | Requires heavy doses of manures, lime, and nitrogenous-phosphatic fertilizers to sustain plantation crops. | Candidates confuse Laterite with Black soil; Laterite has been leached of its bases and silica. |

### 3. Crop Association & Agro-Climatic Match Triggers

| Crop / Vegetation Clue in Stem | Direct Target Soil | Typical Geographic Belt |
|---|---|---|
| **Cotton (कपास)** / "Black Cotton Soil" / Rainfed pulses & oilseeds | **Black Soil / Regur** | Deccan Trap: Maharashtra, Malwa (MP), Saurashtra (GJ), North Karnataka |
| **Zafran / Saffron (केसर / जाफरान)** / Almond & Walnut orchards | **Karewa Soil** *(करेवा)* | Kashmir Valley (Pampore plateau terraces) & Bhadarwah (J&K) |

| Crop / Vegetation Clue in Stem | Direct Target Soil | Typical Geographic Belt |
|---|---|---|
| **Cashew nut (काजू)**, Tapioca, Tea, Coffee, Rubber, Cinchona | **Laterite Soil** *(लेटराइट)* | Western Ghats slopes (Malabar/Kerala, Coorg), Meghalaya hills |
| **Wheat, Rice, Sugarcane, Jute, Mustard** (Intensive foodgrain bowl) | **Alluvial Soil** *(जलोढ़)* | Indo-Gangetic Plains (UP, Bihar, WB, Punjab, Haryana) |
| **Bajra, Jowar, Ragi, Pulses, Castor, Groundnut** (Rainfed coarse crops) | **Red Soil** *(लाल मृदा)* | Rayalaseema (AP), Tamil Nadu plains, Chotanagpur plateau, Odisha |
| **Below sea-level paddy farming** in waterlogged backwater polders | **Kuttanad Kari Soil** | Alappuzha & Kottayam districts (Kerala) |

### 4. Regional Soil Nomenclature Triggers (UP-PCS Special)

| Local Nomenclature / Clue Phrase | Exact Regional Setting | Associated Soil Class & Properties |
|---|---|---|
| **Mar (मार)** | Bundelkhand (Jalaun, Jhansi, Hamirpur) | Deep black clayey regur variant; extremely sticky, highly fertile for gram and wheat. |
| **Kabar (काबर)** | Bundelkhand (Banda, Hamirpur, Mahoba) | Coarser black soil variant; sets into rock-hard clods upon drying; plowing must be timed precisely. |
| **Parua / Padwa (परुआ)** | Bundelkhand (Hamirpur, Jalaun, Yamuna ravines) | Light reddish/yellowish sandy loam; responds well to light irrigation and fertilizers. |
| **Rakar (राकर)** | Bundelkhand (Sloping edges, ravine fringes, hill tops) | Shallow, gravelly, eroded red soil; low water retention; suitable only for *bajra* or sesamum. |
| **Bhonta / Monta (भोंटा)** | Vindhyan hill tracts (Mirzapur, Sonbhadra, Prayagraj trans-Yamuna) | Coarse, stony, reddish sandstone-derived soil; low fertility, rainfed crops. |
| **Bhabar (भाबर)** | Sub-Himalayan foothill strip (northern UP / Uttarakhand border) | Porous gravel-and-boulder belt; surface water sinks and flows underground; unsuited for crops. |
| **Tarai (तराई)** | South of Bhabar strip | Damp, waterlogged silty belt; underground streams re-emerge; high nitrogen & organic matter; sugarcane and paddy belt. |
| **Bhur (भूर)** | Ganga-Yamuna Doab elevated banks | Undulating, elevated sandy mounds formed by wind-blown sand deposits during dry summer months. |
| **Matasi (मटासी)** | Chhattisgarh plains (Mahanadi Basin) | Yellowish-red sandy-loam soil on drained higher sites; staple for rainfed rice. |

### PYQ — UPPCS Prelims 2018, Q51

Which of the following is concerned with soil?

A. Edaphic

B. Climatic

C. Biotic

D. Topography

**Show answer**

**Ans: A** — Edaphic = the **soil** factor. Climatic, biotic and topography are other site factors, not the meaning of edaphic.

## N.1A World soil belts · phytotypes

### World Zonal Soils (Dokuchaev Tradition)

**The Core Zonal Law:** Soils are not merely broken rock fragments. **Vasily Dokuchaev** (the Russian father of modern pedology) established that over time, **Macro-Climate (temperature + precipitation) and Natural Vegetation Biomes override parent geology**. As you move latitudinally across the globe, climate belts create specific vegetation biomes, which directly generate predictable **Zonal Soil Belts**:

| Soil Type | Etymology & Literal Meaning | Climate Belt & Biome | Pedological Formation & Characteristics | Classic Prelims Match / Key Trap |
|---|---|---|---|---|
| **Chernozem** *(चेरनोज़ेम)* | Russian: *Cherniy* (Black) + *Zemlya* (Earth) = **"Black Earth"** | **Temperate Grasslands** (Eurasian Steppes of Ukraine/Russia, North American Prairies) | Annual death of dense fibrous grass roots; cold winters slow bacterial decay → colossal accumulation of **dark humus (up to 10–16%)**; rich in calcium and moisture. | **The world's most fertile natural soil.** Known as the global "Breadbasket of the World" (wheat belt). |
| **Chestnut** *(चेस्टनट / भूरी स्टेपी मृदा)* | Color of chestnut nut (**Dark brown**) | **Semi-Arid Steppe Margins** (drier edge of Chernozem toward deserts) | Similar to Chernozem but receives **lower rainfall** → lower grass biomass → less humus. The color lightens from pitch black to chestnut brown. | Transitional buffer between humid Chernozem and dry Sierozem. |
| **Prairie** *(प्रेयरी / Brunizem)* | French: *Prairie* = **Meadow** | **Humid Temperate Grasslands** (US Midwest Corn & Soybean Belt) | Higher rainfall than Chernozem; dense tall-grass growth yields rich humus, but higher moisture causes mild base leaching. | Transitional between Chernozem (grassland) and Podzol (forest). Highly productive. |

| Soil Type | Etymology & Literal Meaning | Climate Belt & Biome | Pedological Formation & Characteristics | Classic Prelims Match / Key Trap |
|---|---|---|---|---|
| **Podzol** *(पोडज़ोल)* | Russian: *Pod* (under) + *Zola* (ash) = **"Under-ash"** | **Cold Temperate Coniferous (Taiga / Boreal)** forests (Canada, Siberia, Scandinavia) | Acidic pine needles + heavy snowmelt and organic acids leach out iron and humus downward, leaving a **bleached, ash-grey silica layer** (E-horizon). Strongly acidic and nutrient-poor. | **Acidic forest soil.** Classic Prelims pairing: Taiga/Coniferous woodland + ash-grey leached soil. |
| **Spodosol** *(स्पोडोसोल)* | Greek: *Spodos* (wood ash) | Humid cold temperate climates | **USDA Soil Taxonomy equivalent** for the Podzol family. | Treat Spodosol and Podzol as synonymous in civil services papers. |
| **Laterite** *(लेटराइट)* | Latin: *Later* = **Brick** | **Hot-Wet Tropical / Monsoon** belts with alternating wet and dry seasons | Heavy monsoonal downpours wash away silica and bases (*desilication/eluviation*), leaving an insoluble crust of **Iron (Fe) and Aluminium (Al) oxides**. Bakes rock-hard when dry. | **Intensely leached & acidic.** Poor for cereal grains; ideal for **cashew, tapioca, tea, coffee, rubber, cinchona**. |
| **Sierozem** *(सिरोज़ेम)* | Russian: *Seriy* (Grey) + *Zemlya* (Earth) = **"Grey Earth"** | **Mid-Latitude Temperate Deserts** and arid scrub (Central Asia, Great Basin) | Rainfall is so low that plant cover is sparse; negligible humus accumulation → pale ash-grey tint. High evaporation draws lime/calcium salts to the surface. | High in mineral salts/lime (*pedocal*), but severely deficient in organic humus and water. |
| **Terra Rossa** *(टेरा रोसा)* | Italian: *Terra* (Earth) + *Rossa* (Red) = **"Red Earth"** | **Mediterranean Karst Terrain** (Italy, Greece, Spain, Croatia) | Formed exclusively on **Limestone / Dolomite** bedrock. Rainwater dissolves and carries away soluble $CaCO_3$, leaving behind an insoluble residual **red clay mantle** enriched with ferric iron. | **High-Yield Prelims Pair:** Limestone weathering + Mediterranean climate. |

### Dokuchaev Memory Shortcuts for Fast MCQ Solving

- **Trick 1: The Russian Color Palette**
- **Cherno** = **Black** → *Black Earth* → Humus-rich Steppe Grassland (Wheat basket).
- **Siero** = **Grey** (सलेटी/धूसर) → *Grey Earth* → Arid Mid-latitude Desert.
- **Pod-zol** = **Ash** (राख) → *Ash-grey leached* → Cold Taiga / Coniferous forest.
- **Trick 2: The Grassland Moisture Gradient (From Wet → Dry) Prairie** (humid, dark brown) → **Chernozem** (ideal, pitch black) → **Chestnut** (drier, brown) → **Sierozem** (desert, grey) *(Mnemonic: Perfect Corn Cheers Sand → Prairie → Chernozem → Chestnut → Sierozem)*
- **Trick 3: 2-Word Fast Match Keys**
- **Terra Rossa ↔ Limestone (Karst) + Mediterranean** *(Italian red sauce on Mediterranean limestone).*
- **Laterite ↔ Tropical Monsoon Leaching + Brick**.

### Soil Water Dynamics & Capillary Rise

- **Capillary Rise (केशिका क्रिया):** Smaller pore spaces generate significantly stronger capillary suction climb. Effectiveness order: **Clay** > **Loam / Silt** > **Sandy soil** > **Sand**
- **Water-Holding Capacity (जल धारण क्षमता):** Governed by fine particle surface area and clay mineral lattice. Capacity order: **Clay** > **Silt** > **Sand**

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### Phytotypes ↔ Ecological Substrate (Plant Habitat Classification)

Every term ends with **-phyte** (Greek *phyton* = **plant**). The prefix is a standard science root word describing the environment:

| Plant Label (पादप वर्ग) | Prefix Meaning | Everyday Science Anchor Word | Natural Habitat & Substrate | Typical Examples |
|---|---|---|---|---|
| **Halophyte** *(लवणमृद्भिद)* | **Halo** = Salt | **Halogen** *(Chlorine/Iodine form salts like NaCl)*; **Halite** *(rock salt)* | **Saline soils / Coastal salt marshes** *(लवणीय मृदा)* | Mangroves (*Rhizophora*, Sundari), *Salicornia* |
| **Cryophyte** *(शीतमृद्भिद)* | **Cryo** = Extreme Cold / Ice | **Cryogenic Engine** *(ISRO's liquid $H_2/O_2$ rocket engine)*; **Cryotherapy** *(ice therapy)* | **Cold, Polar, Tundra & Snowfields** *(शीत / टुंड्रा क्षेत्र)* | Snow mosses, alpine lichens |
| **Lithophyte** *(शैलोद्भिद)* | **Litho** = Stone / Rock | **Lithosphere** *(rocky crust)*; **Paleolithic** *(Old Stone Age)* | **Bare Rock Surfaces & Cliffs** *(चट्टानी धरातल)* | Rock-dwelling lichens, ferns, orchids |
| **Xerophyte** *(मरुद्भिद)* | **Xero** = Dry | **Xerox** *(literally "dry copying" without liquid ink)*; **Xerophthalmia** *(dry eyes)* | **Tropical Deserts & Arid Regimes** *(शुष्क मरुस्थल)* | Cactus (*Opuntia*), Acacia (*Babool*), Khejri |
| **Hydrophyte** *(जलोद्भिद)* | **Hydro** = Water | **Hydration**, **Hydroelectric**, **Hydrogen** | **Open Water / Aquatic** *(floating or submerged)* | Lotus, Water Lily, *Hydrilla*, *Eichhornia* |
| **Hygrophyte** *(आर्द्रतोद्भिद)* | **Hygro** = Moisture / Humidity | **Hygrometer** *(instrument to measure air humidity)* | **Swamps, Marshes & High Humidity** *(दलदली / नम स्थल)* | Marsh grasses, wetland ferns, mosses |
| **Psammophyte** *(बालुकोद्भिद)* | **Psammo** = Sand | *Psammology (study of sand)* | **Shifting Sand Dunes & Sand Fields** *(रेतीले टीले)* | Sand binders, marram grass |
| **Chasmophyte** *(दरारोद्भिद)* | **Chasmo** = Fissure / Crevice | *Chasm (deep gorge/cleft)* | **Rock Crevices & Fissures** *(चट्टानों की दरारें)* | Saxifrage, cliff crevicular flora |
| **Oxylophyte** *(अम्लोद्भिद)* | **Oxy** = Acidic | *Oxygen (historically "acid former")* | **Strongly Acidic Soils** *(अम्लीय मृदा)* | Heather, peat moss (*Sphagnum*) |

| Plant Label (पादप वर्ग) | Prefix Meaning | Everyday Science Anchor Word | Natural Habitat & Substrate | Typical Examples |
|---|---|---|---|---|
| **Calciphyte** *(चूनामृद्भिद)* | **Calci** = Lime / Calcium | **Calcium Carbonate** ($CaCO_3$), *Calcification* | **Lime & Calcium-rich Soils** *(चूनायुक्त मृदा)* | Calciphilous limestone flora |

**The Commission's Favorite Distractor:** `Hydrophyte` **vs** `Hygrophyte`

- **HYDRO-phyte → Directly in Open Water (Aquatic)**: The plant actually lives submerged in or floating on a body of water (think: staying **hydrated** in a swimming pool → Lotus).
- **HYGRO-phyte → Swamps & High Atmospheric Humidity**: The plant grows on saturated marshy soil where air humidity is near 100% (think: a **hygrometer** measuring humidity → swamp ferns).

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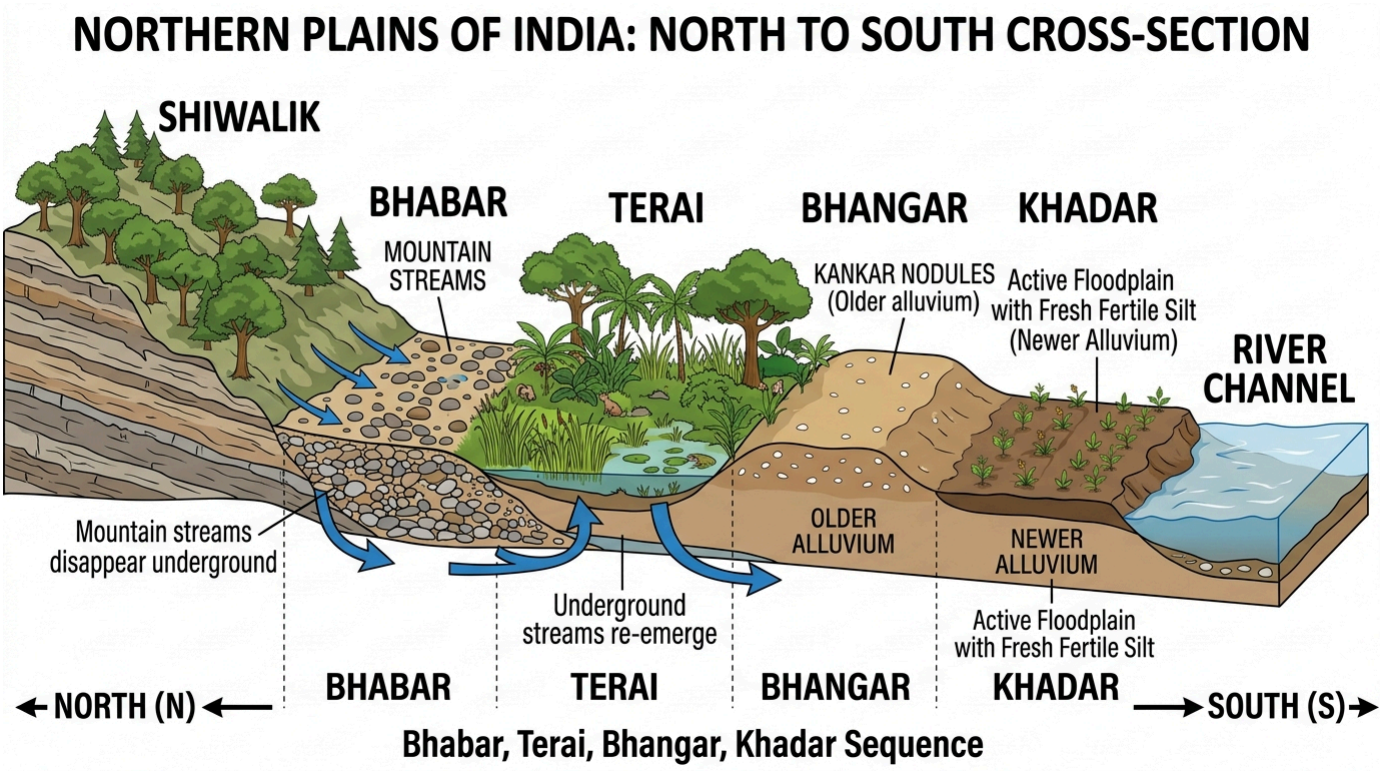
### Soil Ecology & Conservation Highlights

- **Earthworms as "Ecosystem Engineers" (पारिस्थितिकी तंत्र के इंजीनियर):** Earthworms burrow continuously, creating subterranean macropores for aerating roots and water infiltration. Their organic castings enrich topsoil with nitrogen, phosphorus, and microbial flora while loosening heavy clays. They are universally **beneficial** for agriculture (classic wrong option: calling earthworms pests).
- **Tropical Rainforest Paradox:** Equatorial rainforests experience the world's most lush vegetation, yet their soils are naturally **low in available nutrients** because year-round torrential rainfall causes relentless **leaching**, and high heat causes instant microbial oxidation of organic matter before a humus layer can accumulate.
- **Contour Bunding & Terracing (समोच्च रेखीय मेड़बंदी):** Designed specifically to intercept runoff and conserve topsoil on **steep hill slopes** — never pick desert margins, flood plains, or scrublands in IAS/UPPCS MCQs.

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## N.2 Soil Types

### Alluvial




India's **most extensive** soils. NCERT Class 11 uses about **40%**. Other standard notes often write about **46%** (about 15 lakh km²) and say these lands feed more than **40%** of India's people. Keep both figures; do not treat the ~46% figure as a second NCERT value.

- Himalayan rocks are the parent **source**. The soil itself is **transported**.
- Coastal belts also get some alluvium from **wave** action.
- Colour runs light grey to ash grey.
- Profiles are **immature and weak** because the deposit is young.
- Texture is often **loamy and porous**, so drainage is good. Sandy and clayey patches are common. Pebbly beds are rare except in fans.
- **Kankar** beds sit along some river terraces.
- Floods **replenish** silt. That is why khadar stays fertile.
- Texture grades from sandy loam to clay loam with distance from the channel.
- Through a **narrow corridor in Rajasthan** they extend into the plains of **Gujarat**.
- Also in east-coast deltas (**deltaic alluvium**) and Narmada (नर्मदा)–Tapi (तापी) / north Gujarat valleys.
- **Sand content decreases from west to east**; lower–middle Ganga (गंगा) and Brahmaputra are more loamy–clayey.
- These soils back wheat–rice–sugarcane–jute–tobacco–cotton–maize–oilseed belts. They take canal and tubewell irrigation well.
- UP is **overwhelmingly alluvial** — not a Deccan black-soil state.

| | Khadar (newer) | Bangar / Bhangar (भांगर) (older) |
|---|---|---|
| Site | Active floodplain | Higher terraces / older plain |
| Floods | Frequent silt renewal | Rare |
| Texture | Finer silt–clay | Coarser |
| **Kankar** (calcareous nodules) | Possible, usually **fewer** | **Common / typical key fact** |
| Fertility | Generally **higher** | Slightly lower |

- NCERT notes both may contain kankar; stems still treat **bangar** as the **kankar-heavy** terrace.
- **Bhur** is sandy alluvial mounds or ridges along western Gangetic courses (doab).
- Deltaic alluvium on the east coast is finer and more clayey.
- Coastal alluvium may turn saline.

| Chemistry | Status |
|---|---|
| Potash | Generally **rich** (Class 11 fact) |
| Phosphorus | **Poor** in NCERT Class 11. Some other notes say phosphoric acid is “adequate.” Keep the **NCERT: poor P** fact. |
| Lime / alkalies | Variable; often adequate |
| Nitrogen, humus | Generally **low** |

- Bhabar is about **8–16 km** of merged **alluvial fans** along the Shiwalik (शिवालिक) foot. Pebbles make it so porous that streams **vanish**. Only deep-rooted trees do well. It is a **poor** crop belt.
- Terai (तराई) is about **15–30 km** south of Bhabar. Streams **re-emerge**. The belt is marshy, N- and OM-rich, and **phosphate-poor**. Wheat, rice, sugarcane and jute take it once drained. Forest here is wildlife country.
- Bhangar terraces sit about **30 m** above flood level. They are older, often darker and more clayey, with **kankar** a few metres down.
- Khadar is newer flood silt. It is less calcareous than bangar. A fresh layer arrives almost every flood.
- **Rarh** (West Bengal / parts of Jharkhand–Odisha fringe) is older alluvial/lateritic upland south of the Ganga floodplain — do not swap it with khadar or bangar of the active plain.
- Loam is often taught as sand + silt + clay together (option “all types”). Exact percentages vary by book; do not invent a single “official” %.

| Belt | Region | Star crops |
|---|---|---|
| Upper Ganga | Punjab, Haryana, W-UP | Wheat, sugarcane, rice |
| Middle–lower Ganga | E-UP, Bihar, WB | Rice, jute, pulses |

| Belt | Region | Star crops |
|---|---|---|
| Brahmaputra | Assam | Rice; tea on nearby terraces |
| East-coast deltas | Mahanadi (महानदी), Godavari (गोदावरी), Krishna (कृष्णा), Cauvery | Rice, coconut fringe |
| Peninsular valleys | Narmada, Tapi flats; N Gujarat | Mixed food crops |

Canal irrigation raised yields but created local **usar/reh** where drainage failed.

### Black (Regur / cotton soil)

Residual clay on **Deccan Trap basalt** (fissure volcanic / lava rock), including the **Rajmahal** trap. Local names: **regur**, **black cotton soil**, **Tropical Black Earth**, **Tropical Chernozem**.

- In **Tamil Nadu**, gneiss and schist can also parent black soil. Trap black is usually **deep**. Gneiss black is often **shallow**.
- This is a soil of **high temperature and low rainfall** — a dry-hot peninsular group, not a Himalayan group.
- Clay is often **62% or more**. The expanding clay is **montmorillonite**.
- Sticky when wet (hard to till in rain) and cracked when dry. That cracking gives natural **self-ploughing** and oxygenation of lower layers.
- Highly **moisture-retentive**, so rainfed crops can survive dry spells with little irrigation.
- **Upland** black is poorer. **Valley** black is very fertile.
- Dark colour comes mainly from **titaniferous magnetite**, not only humus. Tints run from deep black through medium and shallow shades to mixed red-black.
- **Krebs (Krabs)** treats regur as essentially a **mature** soil (relief + climate).
- The usual area figure is about **5.46 lakh km²**. The rival figure "**46 lakh km²** (16.6%)" cannot both be true. Keep **about 16.6%** / the 5.46-lakh class figure; discard the 46-lakh figure.
- **Jharkhand** black patches are classically in the **Rajmahal** hill belt (not Damodar as the default).

| Chemistry | Status |
|---|---|
| Lime, iron, magnesia, alumina | Rich |
| Potash | Present / variable |
| Nitrogen, phosphorus, organic matter / humus | **Deficient** — so "rich in humus" A/R reasons are **false** |
| Region | Note |
| -------- | ------ |
| Maharashtra (Vidarbha) | Classic / most widespread cotton regur |
| MP / Malwa | Deep black / "lava soil" plateau |
| Gujarat | Black tracts (Saurashtra–Malwa link) |

| Chemistry | Status |
|---|---|
| Telangana / Andhra | Trap extensions; deep in upper Godavari–Krishna |
| N Karnataka | Cotton–jowar (ज्वार) |
| Parts of Tamil Nadu | Local patches (often on gneiss/schist too) |
| Rajmahal (Jharkhand) | Trap black |
| **Himalaya** (हिमालय) | **Not** a black-soil region |

- The star crop is **cotton**.
- Other crops include jowar, wheat, linseed, castor, Virginia tobacco, sunflower and millets.
- Rice and sugarcane grow where irrigation is available.
- Farmers have grown crops on it for centuries with little manure. That is moisture and clay, not a licence to skip fertiliser forever.
- UP black soil is mainly Bundelkhand **Mar/Kabar**, not statewide.

### Red & Yellow

Form on ancient **crystalline igneous and metamorphic** rocks (acid granite, gneiss, quartzite), mainly under **low-rainfall** eastern and southern Deccan conditions.

- Some older notes once called red soils “the largest group.” That is **wrong**. **Alluvial** is the largest. Red is the big peninsular residual family.
- Red colour comes from **ferric oxides** / Fe-oxide **diffusion** through the profile. The iron percentage need not be high.
- **Yellow** soil has more **hydrated** Fe oxides. It is the same family, not a separate ICAR class.
- Texture runs sand to clay. Most patches are **loams**. They **cannot hold water like black clay**.
- **Upland** red is gravelly, porous and poor. **Lower-slope** red is deeper, darker and more fertile.
- They are often **acidic**, poor in lime, magnesia, phosphate, nitrogen and humus, and **fairly rich in potash**.
- Long red-loam stretch along the **piedmont of the Western Ghats**.
- Also in Odisha–Chhattisgarh–Chota Nagpur, almost the whole of **Tamil Nadu**, the **Aravallis**, eastern Rajasthan (**Mewar** (मेवाड़)), and southern parts of the Middle Ganga plain (south Bihar, WB, UP).
- Phosphorus fixation and low OM are the usual management headaches.
- Red soil is **not** automatically laterite.

| Chemistry | Status |
|---|---|
| Iron oxides (colour); potash often adequate | Present |
| N, P, humus; sometimes lime | Often deficient |
| Region | Note |
| -------- | ------ |
| Tamil Nadu, Karnataka, AP/TG | Wide red loams |

| Chemistry | Status |
|---|---|
| Western Ghat piedmont | Red loamy belt |
| Southern Maharashtra | Transition with black soils |
| Odisha, Chota Nagpur, Chhattisgarh | Upland red |
| Parts of MP / NE / S Middle Ganga | Local red–yellow |

Crops: millets, pulses, tobacco, oilseeds, potatoes, fruits; cotton **with irrigation**.

| Feature | Red / Yellow | Laterite |
|---|---|---|
| Parent / climate | Crystalline rocks, often lower rain | Hot-wet **intense leaching** |
| Colour cause | Fe-oxide diffusion | Fe/Al residue after bases/silica leave |
| Bases | Variable | Strongly depleted |
| Typical crops | Millets, pulses | Cashew, tea, coffee, plantations |

### Laterite

Needs **high temperature + heavy rainfall** with **alternate wet and dry** periods. Laterite is treated as an **end-product** of weathering: it will not weather much further, so cut blocks stay durable.

- The name comes from Latin ***later***, meaning **brick**. **F. Buchanan** first described laterite in India (**1807**).
- **Laterisation** washes out **lime and silica**.
- **Iron oxide, aluminium and often bauxite** remain — “poor in Fe/Al” is **false**.
- Potash, nitrogen, lime and magnesia are **poor**. Phosphate may be high as **iron phosphate**. Wetter spots may hold more humus.
- Rapid bacterial decay keeps humus low on most laterite uplands.
- Soft when wet; it **hardens on exposure** into building **bricks**. Blocks can be cut as cakes and then set like iron.
- Field crops need heavy manure; forest clearing speeds erosion on already base-poor profiles.
- **Red laterite** of Tamil Nadu, Andhra Pradesh and Kerala suits tree crops like **cashew** and **tapioca**. Tea, coffee, rubber, cinchona, coconut and arecanut are the plantation set. Some tracts are only grazing or scrub.
- Continuous laterite sits on **Western Ghat summits about 1000–1500 m**, Eastern Ghats (पूर्वी घाट), **Rajmahal**, **Vindhya (विंध्य)–Satpura (सतपुड़ा)–Malwa**, south Maharashtra and parts of Karnataka. Area figures vary — about **1.8 lakh km²** in some notes and about **2.48 lakh km²** in others.
- Among states, laterite is classically strongest in **Kerala (Malabar coast)** then **Maharashtra**. It is **not** the soil of Rajasthan or the UP plain.
- Upper slopes are often more acidic than lower ones (height vs composition fact in some notes).

| Chemistry | Status |
|---|---|
| Iron oxide, aluminium compounds | Excess / rich |
| OM, N, phosphate, calcium, lime, silica | Poor / leached |
| Item | Note |
| ------ | ------ |
| Climate | High T + heavy rain + wet–dry |
| Process | Intense leaching |
| Building use | Bricks |
| Crops | **Cashew, tapioca, tea, coffee, rubber, cinchona** |
| Belt | W Ghats / E Ghats summits, **Rajmahal**, **Vindhya–Satpura–Malwa**, Odisha, WB, Kerala, Karnataka, TN, Assam–Meghalaya hills |
| Not typical | Rajasthan, Uttar Pradesh plains |

### Desert / arid

Core belt is western **Rajasthan (Thar)**; arid fringes touch Gujarat, Punjab and Haryana. Area is about **1.42 lakh km² (~4.3%)**.

- Texture is about **90–95% aeolian sand** and **5–10% clay**. Colour is **red to brown**.
- The sand is largely blown from the **Indus basin and the coast** by **south-west** winds. Neighbouring soils can be **desertified** by dune advance.
- Sandy soils **without clay** also line coasts of Odisha, Tamil Nadu and Kerala. Do not call every coastal sand “Thar.”
- Often **saline** where saline water evaporates into salt pans.
- High temperature plus evaporation leave **moisture and humus** low. Colour stays light.
- **Nitrogen is insufficient**, but some N is present as **nitrates**. **Phosphate is often as high as in alluvium.**
- Calcium **increases downward**. The subsoil may hold about **ten times** the surface calcium (kankar / lime).
- That kankar layer can restrict downward water loss — irrigation moisture then stays usable longer.
- With moisture: barley, cotton, millets, maize and pulses (drought- and salt-tolerant). IGC and wells can reclaim patches **if drainage prevents new salinity**.
- Shelter belts + agro-forestry are the NCERT dune answer.
- These soils are not “forever barren.”

### Mountain / forest

These soils occupy about **2.85 lakh km² (~8.67%)**. They are **heterogeneous**. Two nearby slopes can look different because parent rock, slope and climate all change.

- Formation is governed by **forest litter**. Character **changes with altitude**.
- In the Himalaya they sit mainly in **valleys** and on **less-steep north-facing** slopes. **South-facing** slopes are often too steep and stripped, so soil barely forms.
- Forest soils also occur on the **Western and Eastern Ghats**.
- They are **rich in humus** and deficient in **potash, phosphorus and lime**. They need fertiliser for high yield.
- Steep cold slopes carry thin, immature, skeletal or stony soils.
- Peninsular forest soils take **tea, coffee, spices and tropical fruits**. Himalayan forest soils take **wheat, maize, barley and temperate fruits**.
- **Tea** prefers **slightly acidic**, calcium-poor, well-drained soil (hill slopes or drained valleys) — not alkaline usar and not "any black soil."
- Some older Assertion–Reason stems treated "Himalayan soils are rich in humus" as **false** while keeping "Himalayas have large forest cover" true. For teaching, keep **forest soils as humus-rich** (NCERT and standard notes). Treat that A/R as a paper-specific trap, not a reason to delete the forest-humus fact.
- This belt is **never** the Deccan black-soil region.

### Peaty / marshy

Humid **waterlogging** slows decay. That builds a dark organic pile-up with a lot of soluble salts.

- Core tracts: Kerala **kari** of **Kottayam and Alappuzha** (Kuttanad), Sundarbans fringes, coastal Odisha–Tamil Nadu, **northern Bihar**, **Almora** (अल्मोड़ा) in Uttarakhand.
- Heavy, poorly drained; often **highly acidic**.
- Deficient in **potash and phosphate**.
- High OM is not ready field fertility until **drainage + aeration**.
- Many kari fields stay under water in the rains and go under **paddy** when the rains stop.
- Black colour here is **not** Deccan **regur**.

### Karewa (Kashmir)

- **Karewa** soils are lacustrine (lake-bed) deposits of the Kashmir Valley (also Bhadarwah).
- Fine silt–clay with gravel beds; fossil-bearing in places.
- Classic crops: **saffron**, almond, apple, walnut.
- Do not confuse with ordinary Gangetic alluvium.

### Saline (Usara / Reh)

NCERT name **Usara**. Excess soluble **Na, K, Mg** salts make ordinary crops fail.

- Dry climate + **poor drainage** (swampy tracts also turn saline).
- Capillary rise in irrigated plains plus evaporation leaves a white **reh** crust.
- Structure is sandy to loamy; typically lack **N and Ca**.
- Where: arid/semi-arid belts, canal commands (Punjab, Haryana, W-UP), western **Gujarat**, **eastern-coast / Sundarbans** fringe.

- In the **Rann of Kutch**, the **southwest monsoon** brings salt particles and dumps them as a crust; seawater intrusion also salinises deltas.
- Around the **Gulf of Khambhat**, tides dump salt on the mouths of the **Narmada, Tapi, Mahi (माही) and Sabarmati (साबरमती)**.
- Storm-surge seawater also spoils low coasts of **Andhra Pradesh and Tamil Nadu**.
- Area under saline–alkaline soils is about **68,000 km²**.
- **Indira Gandhi Canal** (इंदिरा गांधी नहर) turned sand into grain land and also **created new usar** where drainage failed.
- Green Revolution (हरित क्रांति) irrigation without outlets **creates** usar.
- Reclamation means **leach salts plus drainage**. More canal water alone worsens the problem. Line canals. Use gypsum on alkali. Salt-tolerant crops (cotton, barley) are only a stop-gap.
- Salinization after irrigation water evaporates can leave salts that make some soils **impermeable** (IAS classic effect centre).
- Among states, **Gujarat** is the usual key for the **largest salt-affected / saline** area. Older stems that put **maximum alkali** in **UP** need the atlas year on the paper.
- **Usar** may support salt-tolerant **paddy** varieties after reclaim; **lime does not reclaim usar** — lime is for acid soils.

### Alkaline (sodic / Kallar)

Excess **exchangeable sodium**, typically **pH > 8.5**.

- Sodium **disperses clay**. Structure collapses and infiltration becomes poor.
- Often mixed with saline usar/kallar in UP–Punjab–Haryana.
- Remedy: **gypsum ($CaSO_4$)** so Ca replaces Na, **then** leach + drain.
- Green manure, salt-tolerant crops, and bunding help.
- **Lime is for acid soils — never swap with gypsum for usar.**

| Feature | Saline | Alkaline (sodic) |
|---|---|---|
| Main problem | Soluble salts | Exchangeable Na |
| pH | Near-neutral to moderately high | Typically **high** |
| Structure | Variable | Dispersed, poor |
| Surface clue | White **reh** | Often darkish alkali stains |
| Key remedy | Leach + drain | **Gypsum** + leach |

### Problem soils (cause and remedy)

| Problem | Cause | Crop effect | Remedy |
|---|---|---|---|
| **Acidic** | Base leaching, high rain (laterite, NE, E Himalayan foothills) | Nutrient fact-up, Al toxicity | **Lime** |
| **Saline** | Soluble salts, capillary rise | Germination failure | **Leaching + drainage** |

| Problem | Cause | Crop effect | Remedy |
|---|---|---|---|
| **Alkaline** | Exchangeable Na, high pH | Structure collapse | **Gypsum + leaching** |
| Waterlogged | High water table / canal seepage | Root oxygen stress | Drainage; lined canals |
| Usar mix | Saline + sodic together | Barren patches | Gypsum + leach + drain + organics |

### Indian soils in one glance

- Peninsula soils are **older and more mature** than the young northern alluvium.
- Most Indian soils are **deficient in nitrogen, humus and mineral salts** after centuries of cropping.
- Plains and valleys are **thick**. Hills and plateau edges are **thin**.
- Alluvial and black soils are the fertile workhorses. Laterite, desert and alkali soils need extra work.
- problems: erosion (Himalaya, Chambal), fertility drop (red/laterite), desertification (मरुस्थलीकरण) (Thar and rain-shadow), waterlogging (Punjab–Haryana), salinity (canal belts including IGC), wasteland and urban (नगरीय) sprawl on farmland.

**PYQ — UPPCS Prelims 2023, Q51**

Which of the following statements is correct about Laterite Soil?

1. Developed in areas with high temperature and heavy rainfall.
2. It is poor in iron oxide and aluminium.

A. Only 1

B. Only 2

C. Both 1 and 2

D. Neither 1 nor 2

**Show answer**

**Ans: A** — (2) is false. Laterite is **rich** in Fe/Al.

**PYQ — UPPCS Prelims 2018, Q31**

Which statement is NOT correct?

A. Black soil is locally called ‘regur’

B. According to Krabs the regur soil is essentially a mature soil

C. Black soils are highly retentive to moisture

D. Black soil is found in Himalayan region

**Show answer**

**Ans: D** — Regur is a Deccan Trap soil. It is not a Himalayan soil.

## N.3 Texture, Fertility, Erosion, Conservation

**Texture** is the share of sand, silt and clay. **Structure** is how particles clump (crumbs versus a sodic mass).

- A clay soil can be productive crumbs (**regur**) or a sterile dispersed sodic mass.
- Fine textures usually hold more cations.
- Coarse textures need more frequent irrigation and OM.

| Texture | Water | Drainage | Tillage | Example |
|---|---|---|---|---|
| Sandy | Low | Fast | Easy | Desert |
| Silty | Medium–high | Medium | Moderate (नरम दल) | Some alluvium |
| Clayey | High | Slow | Hard when wet | Black regur |
| **Loam** | Balanced | Balanced | Favourable | Best farm texture |

| Particle | Diameter |
|---|---|
| **Clay** | **less than 0.002 mm** |
| Silt | About 0.002–0.06 mm |
| Fine sand | About 0.06–2 mm |

| Condition | pH tendency | Remedy hint |
|---|---|---|
| Laterite / NE humid / peaty | Acidic | Lime |
| Usar sodic | Strongly alkaline | Gypsum |
| Healthy alluvial loam | Near neutral | Maintain OM |
| Tea garden ideal | **Slightly acidic** | Avoid stagnant water + lime excess |

- Nutrient availability for many crops peaks near **slightly acid to neutral** (**pH 6–7**).
- P availability falls at both strong acid and strong alkali.
- Among micronutrients Cu, Fe, Mn and Zn, **zinc** is the one most often flagged as **maximum deficient** in Indian soils.
- **Fertility** joins chemical, physical and biological quality. Colour alone never proves it.

| Factor | Role |
|---|---|
| **NPK** (primary macros, not micros) + secondaries/micros | Direct nutrition |
| Humus / OM | Structure, water, cation bank |
| pH | Controls availability |
| Organisms (earthworms, microbes) | Cycling, porosity |

| Factor | Role |
|---|---|
| Moisture & aeration | Root function |
| Drainage | Prevents salinity / waterlogging |

- N is often limiting in Indian soils; legumes + fertilisers help.
- Continuous cropping without OM **mines** nutrients.
- Laterite is short of bases after heavy leaching.
- Desert soil is short of humus.
- Usar / saline–alkali soil is limited by salts.
- Red uplands are often short of nitrogen and humus.
- Earthworms mix/aerate and raise fertility — “friends of farmers.”
- They do **not** have two pairs of hearts; earthworms have **five pairs** of aortic arches.

**Soil Health Card:** nutrient status + fertiliser advice. Launched **19 Feb 2015**, Suratgarh; slogan *Swasth Dharaa, Khet Haraa*.

- **Degradation** is quality decline from salts, exhaustion, compaction or pollution. **Erosion is only one path**.
- Soil degradation is commonly split as physical (erosion), chemical (salinity, pollution) and biological (cover loss).
- Over-irrigation can salinise a flat field with little sheet wash.
- Causes: deforestation, overgrazing, up-down ploughing, mono-cropping, over-irrigation, mining, jhum.
- Water erosion dominates monsoon India; wind dominates the arid west.
- Classic deforestation scars: **Shiwalik chos** of Punjab, **Chambal** (चंबल) ravines.
- Overgrazing in the dry season pulls grass **out by the roots** and pulverises topsoil with hooves.
- Faulty farming includes ploughing **down the slope**, skipping rotation, and unsustainable **jhum** as fallows shrink.

### Soil Erosion Stages (The Water Erosion Ladder)

> **The Water Erosion Sequence (जल अपरदन की क्रमिक अवस्थाएँ): Splash → Sheet → Rill → Gully → Ravine / Badland** *(Mnemonic: Splash Smashes, Sheet Steals, Rill Rakes, Gully Gouges, Ravine Ruins!)*

| Stage / Type | Primary Agent | Physical Mechanism & Diagnostic Feature | Real-World Example & Field Signature | Exam Decision Rule |
|---|---|---|---|---|
| **1. Splash Erosion** *(बूंद अपरदन)* | Falling Raindrops | Kinetic impact of heavy raindrops acts like micro-bombs, smashing soil aggregates (crumbs) and detaching fine particles. | Bare plowed soil during the very first monsoon downpours. | **First / Initiating stage** of water erosion. |
| **2. Sheet Erosion** *(परत अपरदन)* | Surface Runoff (Overland flow) | Thin, uniform wash of topsoil across a broad slope; strips fertile organic humus layer without creating channels. | Gently sloping cultivated farmland after continuous heavy rains. | **"The Silent Thief"** — widespread, devastating, yet goes unnoticed by farmers until crop yields crash. |
| **3. Rill Erosion** *(रिल / अंगुली अपरदन)* | Concentrated Runoff | Runoff gathers into minute, shallow, finger-like incisions (micro-channels) across the field surface. | Sloping fallow fields and orchard floors. | **Key Exam Test:** Small enough that **ordinary tillage/ploughing can easily erase** them. |
| **4. Gully Erosion** *(अवनालिका अपरदन)* | Powerful Channelized Runoff | Rills deepen and widen into steep-sided trenches and gullies that cut through subsoil; fragments farmland. | Plateau margins, foothills, upper catchment of Chambal & Yamuna. | **Key Exam Test:** Deep channels that **normal tillage CANNOT erase**; divides land into unusable fragments. |
| **5. Ravine / Badland** *(बीहड़ / खड्ड भूमि)* | Advanced Networked Gullies | Dense, intricate labyrinth of deep interconnected ravines and canyons; completely strips soil cover down to parent rock. | **Chambal Basin** (Morena, Bhind, Gwalior in MP; Dholpur in RJ; Agra–Etawah–Jalaun in UP). | Classic **Badland Topography** (*उत्खात भूमि*). |
| **Stream Channel Erosion** | Fluvial Streamflow | Under-cutting, scouring, and bank caving along permanent river corridors during peak flood stages. | Floodplains of Ganga, Brahmaputra, Kosi. | Bank erosion and river meandering. |
| **Wind / Deflation** *(वायु अपरदन)* | High-Velocity Wind | Rolling, saltation, and suspension of dry loose sand; dune migration burying productive lands. | Western **Thar Desert** (Rajasthan, southern Haryana, Kutch). | Prevalent in arid zones where vegetative cover is absent. |

### Soil Conservation Methods (Categorized for Effortless Recall)

Instead of a scattered list, conservation methods fall into **five functional domains**:

1. Terrain and slope control

2. Agronomic field methods
3. Wind and arid-zone defences
4. Biological and watershed cover
5. Chemical and hydrological reclamation

### Category 1: Terrain & Slope Engineering (ढाल प्रबंधन)

| Method (विधि) | Core Mechanism (कार्य सिद्धांत) | Ideal Topographic Setting |
|---|---|---|
| **Contour Ploughing** *(समोच्च रेखीय जुताई)* | Ploughing strictly **across (along) contour lines**, horizontal to the slope (**never** up-down). Furrows act as micro-dams that intercept runoff and let water infiltrate. | **Gentle farm slopes** (up to ~6° slope). |
| **Contour / Graded Bunding** *(समोच्च मेड़बंदी)* | Earthen embankments (*bunds*) constructed along contours to impound runoff; graded bunds gently channel excess water safely to drainage outlets. | **Plains and plateau margins** with moderate slopes and permeable soils. |
| **Contour Trenching** *(समोच्च खाइयाँ)* | Excavating continuous or staggered trenches along hill contours to catch surface runoff, arrest silt, and recharge subterranean aquifers. | **Barren hill slopes and forest watersheds**. |
| **Terracing / Bench Terracing** *(सीढ़ीदार खेत)* | Carving steep hillsides into flat, step-like benches bounded by riser walls; transforms rapid downhill torrents into calm, contained irrigation steps. | **Steep mountains & hill tracts** (Himalaya, Western Ghats). *(Trap: Never select flat plains or Thar desert!).* |
| **Check Dams & Gully Plugs** *(रोक बांध / गली प्लग)* | Small rock, brushwood, or concrete barriers erected across gully beds to dissipate water velocity, trap moving silt, and stabilize gully heads. | **Active gullies and ravines** (Chambal, Yamuna ravines). |

### Category 2: Agronomic & Cropping Practices (खेत प्रबंधन)

| Method (विधि) | Core Mechanism (कार्य सिद्धांत) | Ideal Farming Setting |
|---|---|---|
| **Strip Cropping** *(पट्टीदार खेती)* | Alternating strips of erosion-permitting crops (e.g., maize, cotton, jowar) with close-growing cover crops (e.g., grasses, pulses, clover) across the slope. | Open undulating plains and windy agricultural slopes. |
| **Cover Cropping & Mulching** *(आवरण फसलें एवं पलवार)* | Growing dense ground cover crops or spreading crop residue/straw over bare soil; cushions kinetic raindrop impact and halts sheet wash. | Rainfed farms, orchards, and between crop seasons. |
| **Crop Rotation & Mixed Cropping** *(फसल चक्र एवं मिश्रित खेती)* | Alternating tap-rooted and fibrous-rooted crops with leguminous nitrogen fixers; continuously replenishes fertility, preserves crumbs, and avoids seasonal bare exposure. | Intensive cropping systems across India. |
| **Zero / Reduced Tillage + Stubble Retention** *(शून्य जुताई)* | Drilling seeds directly into untilled soil through prior crop stubble; eliminates mechanical soil pulverization and conserves deep soil moisture. | Rainfed rabi crops and intensive rice-wheat tracts. |

| Method (विधि) | Core Mechanism (कार्य सिद्धांत) | Ideal Farming Setting |
|---|---|---|
| **Early-Maturing Crop Varieties** | Planting short-duration cultivars that establish rapid ground canopy and harvest before late-monsoon downpours. | Drought-prone and rainfed dryland belts. |

### Category 3: Wind & Arid Zone Defense (मरुस्थल व वायु अवरोधक)

| Method (विधि) | Core Mechanism (कार्य सिद्धांत) | Ideal Geographical Setting |
|---|---|---|
| **Shelterbelts / Windbreaks** *(रक्षक मेखला / वातरोधी वृक्ष)* | Planting multiple linear rows of dense, deep-rooted trees and shrubs perpendicular to prevailing erosive winds to reduce wind velocity and deflect sandstorms. | **Western Rajasthan (Thar)**, coastal sandy belts, canal banks. |
| **Sand Fences & Dune Stabilization** | Erecting porous brushwood fences and planting xerophytic sand-binders (*Calligonum*, *Cenchrus* grass) to immobilize shifting sand dunes. | Active barchan and longitudinal dune fields. |

### Category 4: Biological & Watershed Recovery (जैविक एवं वानिकी उपाय)

| Method (विधि) | Core Mechanism (कार्य सिद्धांत) | Key Focus Region |
|---|---|---|
| **Afforestation & Reforestation** *(वनीकरण)* | Planting native trees with deep root networks that anchor unstable regolith, encourage leaf-litter formation, and regulate watershed hydrology. | Degraded hill catchments and ravine rims. |
| **Regulated Rotational Grazing** *(नियंत्रित चराई)* | Limiting livestock numbers and rotating grazing pastures to prevent overgrazing, which strips grass roots and pulverizes topsoil into loose dust. | Hill pastures, grassland tracts, and scrub forests. |
| **Agro-forestry** *(कृषि-वानिकी)* | Integrating multipurpose timber, fruit, and fodder trees with agricultural crops and livestock on the same land unit. | Drylands and degraded farm tracts. |
| **Checking Jhum / Shifting Cultivation** | Replacing unsustainable slash-and-burn cycles with permanent settled bench terracing and horticultural plantations. | North-Eastern Hill states and parts of the Eastern Ghats. |

### Category 5: Chemical & Hydrological Reclamation (सुधार उपाय)

| Problem Condition | Scientific Remedy & Action | Typical Focus Tract |
|---|---|---|
| **Alkaline / Sodic Soils (*Usar*)** | Application of **Gypsum ($CaSO_4 \cdot 2H_2O$)** or Iron Pyrites ($FeS_2$) to replace toxic exchangeable $Na^+$ with $Ca^{2+}$. | Canal-irrigated plains of UP, Punjab, Haryana. |
| **Acidic Soils (Low pH)** | Application of **Agricultural Lime ($CaCO_3$)**, Dolomite, or Slaked Lime to neutralize soil acidity and restore base saturation. | Laterite belts (Kerala, Karnataka), humid North-East hills. |
| **Waterlogged & Saline Soils** | Installation of subterranean tile drainage + lining of canals to lower elevated water tables and halt capillary salt deposition. | Command areas of major canal irrigation projects. |

### Key Institutions & Milestones in Soil Conservation

| Initiative / Landmark | Core Fact & Prelims Takeaway |
|---|---|
| **Mitti Bachao Andolan** *(Save the Soil Movement)* | Started in **1977** at **Hoshangabad (now Narmadapuram), Madhya Pradesh**, against waterlogging and soil salinisation caused by the **Tawa Dam Project**. *(Classic Trap: Not Thane, Mysore, or Darbhanga!).* |
| **Central Soil Conservation Board (CSCB)** | Established in **1953** by the Government of India to formulate soil conservation programs across different agro-climatic zones. |
| **World Soil Day** | Celebrated annually on **5 December** (birthday of Thailand's King Bhumibol Adulyadej, who championed soil science). |

- **High-Yield Date Pairings for Prelims Matching:**
- **18 April:** World Heritage Day *(विश्व धरोहर दिवस)*
- **22 May:** International Day for Biological Diversity *(अंतर्राष्ट्रीय जैव विविधता दिवस)*
- **5 June:** World Environment Day *(विश्व पर्यावरण दिवस)*
- **29 July:** International Tiger Day *(अंतर्राष्ट्रीय बाघ दिवस)*
- **5 December:** World Soil Day *(विश्व मृदा दिवस)*

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**PYQ — UPPCS Prelims 2020, Q81**

Where was the Mitti Bachao (Save the Soil) Movement started in India?

A. Thane, Maharashtra

B. Mysore, Karnataka

C. Darbhanga, Bihar

D. Hoshangabad, Madhya Pradesh

**Show answer**

**Ans: D** — Hoshangabad (now Narmadapuram), Madhya Pradesh.

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## N.4 UP Focus

### UP soil belts — expanded teaching map

- **Terai** (तराई) — alluvial, moist, N-rich; rice–wheat after drainage.
- **Middle doab** — khadar/bangar mix; wheat–sugarcane Green Revolution belt.
- **Bundelkhand** — **Mar/Kabar** black and **Parua/Rakar** red; drought-prone plateau.
- **Vindhyan fringe** — **Bhonta/Monta** shallow soils on sandstone.
- **Rohilkhand–Awadh (अवध)** — deep alluvium; mentha and cane on lighter loams.

- **Usar/Reh/Kallar** — saline-alkaline patches in western canal tracts; gypsum and drainage remedies apply.

~**90% alluvial** (khadar / bangar / bhur).

- Do not answer every Uttar Pradesh soil question with “alluvial” when the soil named is Mar, Kabar, Parua, or Bhonta.
- Local names and foothill belts are the frequent UPPCS traps.

| Group | Where | Character | Crop / tag |
|---|---|---|---|
| **Bhabar** | Shivalik (शिवालिक) foothill strip (~8–16 km); Saharanpur–Bijnor west | Shallow, pebbly, porous; streams **disappear** into fans | **Poor** for crops |
| **Tarai** | South of Bhabar (~15–30 km belt) | Streams **re-emerge**; moist, N + OM rich, silty–clayey | Rice, wheat, sugarcane |
| Gangetic alluvium | West–east plains | Deep fertile; khadar/bangar/bhur | ~90% of state |
| Bundelkhand | Jhansi (झांसी), Lalitpur, Hamirpur, Jalaun, Banda, Mahoba, Chitrakoot (चित्रकूट) | Mar/Kabar black; Parua/Rakar red | Local-name favourite |
| Vindhyan / Baghelkhand fringe | Mirzapur (मिर्ज़ापुर), Sonbhadra (सोनभद्र), parts of Prayagraj (प्रयागराज)–Chandauli | Red–yellow with Fe/Al; **Bhonta/Monta** | Pulses, millets, oilseeds |
| Aravali fringe | Parts of Agra | Reddish, Chambal-brought | Local fringe |
| Usar / Reh | W-central canal plains | Saline–alkaline crusts | Gypsum + drainage |
| Local name | Colour family | Trait | Core districts |
| --- | --- | --- | --- |
| **Mar** | Black | Fine clayey and OM-rich | Jalaun, Hamirpur, Jhansi, Banda |
| **Kabar** | Dark to grey-black | Coarse and extremely sticky | Lalitpur, Banda, Jalaun, Hamirpur |
| **Parua** | Yellowish red | Sandy to clayey; well aerated | Throughout Bundelkhand |
| **Rakar (pathari)** | Reddish-brown | Coarse on slopes; slightly acidic | Sloping Bundelkhand |

- **Mar** often waterlogs and carries kankar.
- **Kabar** hardens into blocks when dry.
- **Rakar** is leaching-prone on slopes. | **Bhonta / Monta** | Vindhyan red–yellow | Plateau fringe | Mirzapur–Sonbhadra–Prayagraj | | Reh / Usar / Kallar / Thur / Banjar (बंजर) | Saline–alkaline | Canal tracts | Aligarh (अलीगढ़), Mainpuri, Etawah, Kanpur (कानपुर), Unnao, Raebareli, Lucknow (लखनऊ) belt |

| Bhur | Sandy ridges | Upper Ganga–Yamuna Doab | Western plain | | Jalodh / Cop | Alluvial local names | Plains | — | | Problem | Hotspot | Fix | |---------|---------|-----| | Usar / reh | W-central canal tracts | Gypsum + drainage + OM | | Ravines / gullies | Agra–Etawah–Jalaun | Check dams, afforestation, grazing control | | Wind (summer) | Agra–Mathura (मथुरा)–Etawah | Shelter belts / cover | | Waterlogging | Tarai + canal seepage | Drainage; canal lining | | Fertility decline | Intensive plains | Rotation, legumes, organic returns |